Side	Claim (logical content)	Status
Original argument (BMV logic)	P1: If two masses entangled only via gravitational interaction, the mediator is nonclassical (under the locality assumption)	Holds (premise-dependent)
Original argument	P2: The observation is realizable on the tabletop	Open (cost quantified: $10^9 \sim 10^{21}$ runs)
Counterexample (Aziz-Howl)	P3: A "classical gravity + quantum matter" framework exists that reproduces the entanglement	Holds (within its own framework)
Counterexample	P4: Therefore GIE cannot prove quantization of gravity	Does not hold (an overreach beyond P3′s logical content)
Rebuttal (Marletto et al. + 2026 dissection)	P5: P3′s entanglement sits in the matter sector (correlator vanishes for distinct matter types)	Holds (model-independent restatement)
Rebuttal	P6: The standard BMV inference is fully preserved	Holds (for LOCC-class channels)

Table 1: Table 1: Separation of the six propositions. Key: P4 is an overreach from P3; P6 holds for LOCC-class channels, not for all classical models.

The GIE 2025 Debate, Logically Reconstructed: What Each Side Actually Proved

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Abstract The 2025 debate over gravity-induced entanglement (GIE)—the original argument, the classical-quantum hybrid counterexample, and the matter-sector rebuttal—is commonly summarized as "who was right and who was wrong." This note argues that the summary omits the most valuable content: the three sides proved **logically different propositions**. Using a table and three-model simulations, we separate the three sides' claims into six precise propositions, marking each one's logical content and its unresolved part. Conclusion: the debate's net product is not a verdict but a set of precise statements about "which observation excludes which theory"—precisely how the adjudication-matrix methodology occurs in real scientific practice.**Keywords** gravity-induced entanglement; LOCC; quantum gravity; scientific methodology

Three Sides, Six Propositions

Logical Reconstruction

The correct content of the original argument (P1): under the local-mediator assumption, entanglement generation excludes local classical channels. This is a standard theorem of quantum information; it is not in dispute.

The correct content of the counterexample (P3): a specific classical-quantum hybrid theory can produce the observable. Its value lies in making P1′s **premise** explicit—P1′s range of validity depends on the precise form of the "local mediator" assumption.

The counterexample's overreach (P4): from "there exists a hybrid model that reproduces" to "therefore quantization cannot be proved" ignores two points: the model's matter sector is itself quantum (the P5 dissection); and Newtonian gravity plus classical nonlocal evolution likewise produces entanglement (Marchese et al. 2025), showing that P1$'$s exclusion target was LOCC all along, not "everything classical."

The correct content of the rebuttal (P5, P6): the entanglement channel sits in the matter sector; the standard inference is fully preserved for LOCC-class channels.

Net Product: A Set of Precise Statements

The debate's net product is not a verdict but the establishment of the following statements:

- GIE is a **LOCC excluder**: observing entanglement excludes local classical channels, not all classical models.
- Adjudication targets must be made precise: different exclusion strengths correspond to different model classes (LOCC / semiclassical / hybrid).
- Premises must be made explicit: the local-mediator assumption is an input to P1, not an output.
- The engineering boundary of the proposition consists of the run-cost magnitude ($10^9 \sim 10^{21}$) and the apparatus-recoil constraint (100 μm).

These four are exactly the content of adjudication-matrix rows 6-7. The methodological conclusion: **when three sides of a debate are all partially right, the scientific product is the precise separation of propositions, not a verdict**—a complete occurrence of the arrow discipline in real practice.

Appendix: Three-Model Simulation Summary

Numerical simulation of the four-dimensional Hilbert space implements the exclusion logic: the quantum model gives negativity $N \approx 0.078$ and CHSH violation 0.024; mean-field and LOCC models give exactly zero; the continuous-variable verification converges with deviation $e^{-2\alpha^2}$. The simulation favors no side —it precisely renders P1$'$s range of validity.

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